

Auteur: Zaky Souai
Studierichting: Informatica
Oefeningenreeks 5B nr. 2.7

$$\begin{aligned} & \int_{-\infty}^{+\infty} \frac{x}{(x^2 + 4)^2} dx \\ &= \lim_{a \rightarrow -\infty} \int_a^0 \frac{x}{(x^2 + 4)^2} dx + \lim_{b \rightarrow +\infty} \int_0^b \frac{x}{(x^2 + 4)^2} dx \\ \text{Substitutie: } u &= x^2 + 4 \Rightarrow du = 2x dx \Rightarrow \frac{1}{2} du = x dx \\ \Rightarrow \int \frac{x}{(x^2 + 4)^2} dx &= \frac{1}{2} \int \frac{du}{u^2} = \frac{-1}{2u} + C = \frac{-1}{2(x^2 + 4)} + C \\ \Rightarrow \int_{-\infty}^{+\infty} \frac{x}{(x^2 + 4)^2} dx &= \lim_{a \rightarrow -\infty} \left[\frac{-1}{2(x^2 + 4)} \right]_a^0 + \lim_{b \rightarrow +\infty} \left[\frac{-1}{2(x^2 + 4)} \right]_0^b \\ &= \frac{-1}{8} + \frac{1}{8} = 0 \end{aligned}$$

References